

APPENDIX 12 – RELATIONAL DENSITY SCORING

Table 1 Relational density scoring, Part 1

Case	Org Type	Partner Types Engaged	D1: Partner Diversity (0-12)	D1 Normalized ($\div 12$)	PT Categories Present	D2: Relationship Type Range (0-6)	D2 Normalized ($\div 6$)
CLEAR	DSO	Indigenous Gov (1), Community (2), Academic (3), Digital Rights (10), Social Movement (11)	5	0.42	PT1, PT3, PT4	3	0.5
FNIGC	DSO	Indigenous Gov (1), Academic (3), Government (4), International (5)	4	0.33	PT1, PT2, PT3	3	0.5
GIDA	DSO	Indigenous Gov (1), Academic (3), International (5), Digital Rights (10), Social Movement (11)	5	0.42	PT1, PT2, PT3	3	0.5
Maiaam Nayri Wingara	DSO	Indigenous Gov (1), Academic (3), International (5)	3	0.25	PT1, PT2	2	0.33
Masakhane	DSO	Community (2), Academic (3), Corporate/Tech (7), Digital Rights (10)	4	0.33	PT1, PT4	2	0.33

Monlam AI	DSO	Indigenous Gov (1), Academic (3), Social Movement (11), Cultural/Religious (12)	4	0.33	PT3, PT4	2	0.33
Te Hiku Media	DSO	Indigenous Gov (1), Academic (3), International (5), Digital Rights (10)	4	0.33	PT1, PT3	2	0.33
Te Mana Raraunga	DSO	Indigenous Gov (1), Academic (3), Government (4), International (5)	4	0.33	PT1, PT2, PT3	3	0.5
CoopCycle	DC	Government (4), Cooperative (8), Labor (9)	3	0.25	PT1, PT5	2	0.33
Drivers Cooperative	DC	Philanthropic (6), Cooperative (8), Labor (9), Social Movement (11)	4	0.33	PT1, PT2	2	0.33
Fairbnb	DC	Community (2), Academic (3), Philanthropic (6), Cooperative (8), Social Movement (11), Cultural (12)	6	0.5	PT1, PT5	2	0.33
Groupmuse	DC	Community (2), Philanthropic (6), Cultural/Religious (12)	3	0.25	PT1, PT5	2	0.33
Les Mercedes	DC	Community (2), Government (4), Cooperative (8),	4	0.33	PT3, PT5	2	0.33

		Social Movement (11)					
Loomio	DC	Community (2), Government (4), Cooperative (8), Digital Rights (10), Social Movement (11)	5	0.42	PT1, PT6	2	0.33
MiData	DC	Academic (3), International (5)	2	0.17	PT1, PT4	2	0.33
Namma Yatri	DC	Community (2), Government (4), Corporate/Tech (7), Labor (9), Digital Rights (10)	5	0.42	PT3, PT4, PT5	3	0.5
SOLshare	DC	Community (2), Government (4), Philanthropic (6), Corporate/Tech (7)	4	0.33	PT5	1	0.17
Som Energia	DC	Government (4), International (5), Cooperative (8), Social Movement (11)	4	0.33	PT1, PT2	2	0.33
ARDC	PGO	Academic (3), Government (4), International (5)	3	0.25	PT1, PT5	2	0.33
DECODE	PGO	Academic (3), Government (4), International (5), Philanthropic (6), Corporate/Tech	6	0.5	PT5	1	0.17

		(7), Digital Rights (10)					
DEF	PGO	Community (2), Academic (3), Government (4), International (5), Philanthropic (6), Corporate/Tech (7), Digital Rights (10), Social Movement (11)	8	0.67	PT1, PT2, PT4, PT5	4	0.67
Nea Guinea	PGO	Community (2), Philanthropic (6), Cooperative (8), Digital Rights (10), Social Movement (11)	5	0.42	PT1, PT5	2	0.33
OMF	PGO	Academic (3), Government (4), International (5), Philanthropic (6), Corporate/Tech (7), Social Movement (11)	6	0.5	PT2, PT5	2	0.33
OpenSAFELY	PGO	Academic (3), Government (4), Corporate/Tech (7), Digital Rights (10)	4	0.33	PT4, PT5	2	0.33
Participatory Brazil	PGO	Community (2), Academic (3), Government (4), Digital Rights (10)	4	0.33	PT4, PT5	2	0.33

RDA	PGO	Academic (3), Government (4), International (5), Corporate/Tech (7)	4	0.33	PT1, PT2, PT5	3	0.5
SDI	PGO	Community (2), Government (4), International (5), Social Movement (11)	4	0.33	PT1, PT2	2	0.33
Solar Commons	PGO	Indigenous Gov (1), Community (2), Academic (3), Government (4), Corporate/Tech (7), Cultural (12)	6	0.5	PT3, PT4, PT5	3	0.5
CTC	PPS	Community (2), Digital Rights (10), Social Movement (11)	3	0.25	PT1, PT5, PT6	3	0.5
DAIR	PPS	Community (2), Academic (3), Digital Rights (10), Social Movement (11)	4	0.33	PT1, PT3, PT4	3	0.5
Enspiral	PPS	Community (2), Cooperative (8), Digital Rights (10), Social Movement (11)	4	0.33	PT1, PT6	2	0.33
Guifi.net	PPS	Community (2), Academic (3), Government (4), International (5), Corporate/Tech	6	0.5	PT1, PT2, PT5	3	0.5

		(7), Digital Rights (10)					
Linux Foundation	PPS	Academic (3), Government (4), Corporate/Tech (7), Digital Rights (10)	4	0.33	PT4, PT5	2	0.33
Matrix	PPS	Government (4), Philanthropic (6), Corporate/Tech (7), Digital Rights (10)	4	0.33	PT4, PT5	2	0.33
OpenAgriNet	PPS	Community (2), Academic (3), Government (4), International (5), Corporate/Tech (7), Digital Rights (10)	6	0.5	PT1, PT2, PT4, PT5	4	0.67
OpenStreetMap	PPS	Community (2), Government (4), International (5), Corporate/Tech (7)	4	0.33	PT4, PT5	2	0.33
P2P Foundation	PPS	Community (2), Academic (3), Government (4), International (5), Philanthropic (6), Digital Rights (10), Social Movement (11)	7	0.58	PT1, PT2, PT5	3	0.5
Sensorica	PPS	Community (2), Academic (3), Government (4),	5	0.42	PT1, PT4, PT5	3	0.5

		Corporate/Tech (7), Digital Rights (10)					
Stop LAPD Spying	PPS	Community (2), Academic (3), Digital Rights (10), Social Movement (11)	4	0.33	PT1, PT3	2	0.33
TIC A.C.	PPS	Indigenous Gov (1), Community (2), Digital Rights (10), Social Movement (11)	4	0.33	PT3, PT4, PT5	3	0.5
WAO	PPS	Academic (3), Philanthropic (6), Labor (9), Digital Rights (10)	4	0.33	PT1, PT4, PT5	3	0.5
WikiHouse	PPS	Community (2), Academic (3), Philanthropic (6), Corporate/Tech (7)	4	0.33	PT1, PT5	2	0.33
Zenzeleni	PPS	Indigenous Gov (1), Community (2), Academic (3), Government (4), Philanthropic (6), Corporate/Tech (7), Digital Rights (10)	7	0.58	PT1, PT3, PT4, PT5	4	0.67

Table 2 Relational density scoring. Part 2

Case	Org Type	Network Reach Description	D3: Network Reach (0-3)	D3 Normalized ($\div 3$)	Inter-Case Connection Description	D4: Inter-Case Connectivity (0-3)	D4 Normalized ($\div 3$)	Composite Score (0-4)
CLEAR	DSO	International (GOSH movement) + National (Canada) + Local (Newfoundland)	3	1	No documented connections	0	0	1.92
FNIGC	DSO	International + National + Regional	3	1	Direct: GIDA, TMR, MNW (Indigenous cluster)	3	1	2.83
GIDA	DSO	International + National (4 countries) + Local	3	1	Direct: TMR, MNW, FNIGC (Indigenous cluster)	3	1	2.92
Maïam Nayri Wingara	DSO	International + National + Local	3	1	Direct: GIDA, TMR, FNIGC (Indigenous cluster)	3	1	2.58
Masakhane	DSO	International (global tech) + Pan-African + Local	3	1	No documented connections	0	0	1.66
Monlam AI	DSO	International + National + Local	3	1	No documented connections	0	0	1.66
Te Hiku Media	DSO	International + National + Local (iwi)	3	1	Direct: TMR (Indigenous cluster)	2	0.67	2.33
Te Mana Raraunga	DSO	International + National + Local	3	1	Direct: GIDA, FNIGC, MNW, Te Hiku	3	1	2.83

					(Indigenous cluster)			
CoopCycle	DC	International (Argentina) + National (France) + Local (Paris)	3	1	Shared space: platform cooperative cluster	1	0.33	1.91
Drivers Cooperative	DC	National + Local (NYC)	2	0.67	Shared space: platform cooperative cluster	1	0.33	1.66
Fairbnb	DC	International (Europe) + National + Local	3	1	Shared space: platform cooperative cluster	1	0.33	2.16
Groupmuse	DC	National (US cities) + Local	2	0.67	No documented connections	0	0	1.25
Les Mercedes	DC	Local (Barcelona)	1	0.33	Shared space: platform cooperative cluster	1	0.33	1.32
Loomio	DC	International + National + Local	3	1	Direct: Enspiral (spawned from)	2	0.67	2.42
MiData	DC	International (Europe) + National (Switzerland)	3	1	No documented connections	0	0	1.5
Namma Yatri	DC	National (India) + Local	2	0.67	No documented connections	0	0	1.59
SOLshare	DC	National (Bangladesh) + International (investors/forums)	2	0.67	No documented connections	0	0	1.17
Som Energia	DC	International (Europe) + National	3	1	Shared space: Spanish cooperative	1	0.33	1.99

		(Spain) + Local (Catalonia)			cluster (Guifi.net)			
ARDC	PGO	International + National (Australia)	3	1	Direct: RDA (research data governance)	2	0.67	2.25
DECODE	PGO	International (6 EU countries) + Local (Barcelona, Amsterdam)	3	1	No documented connections	0	0	1.67
DEF	PGO	International + National (India) + Local (rural)	3	1	No documented connections	0	0	2.34
Nea Guinea	PGO	International + National (Greece) + Local (rural)	3	1	Shared space: open knowledge commons cluster	1	0.33	2.08
OMF	PGO	International + National + Local (130+ cities)	3	1	No documented connections	0	0	1.83
OpenSAFELY	PGO	National (UK)	2	0.67	No documented connections	0	0	1.33
Participatory Brazil	PGO	National (Brazil)	2	0.67	No documented connections	0	0	1.33
RDA	PGO	International + National + Regional	3	1	Direct: ARDC (research data governance)	2	0.67	2.5
SDI	PGO	International (Africa, Asia, Latin America) + National + Local	3	1	No documented connections	0	0	1.66

Solar Commons	PGO	National (US, multiple sites) + Local	2	0.67	No documented connections	0	0	1.67
CTC	PPS	International + National + Local (Detroit)	3	1	Shared space: community network builders cluster	1	0.33	2.08
DAIR	PPS	International + National	3	1	Shared space: algorithmic accountability cluster	1	0.33	2.16
Enspiral	PPS	International + National (NZ) + Local	3	1	Direct: Loomio (spawned)	2	0.67	2.33
Guifi.net	PPS	International (EU) + National + Local	3	1	Shared space: community networks + Spanish cooperative cluster	1	0.33	2.33
Linux Foundation	PPS	International + National	3	1	No documented connections	0	0	1.66
Matrix	PPS	International + National	3	1	No documented connections	0	0	1.66
OpenAgriNet	PPS	International (multi-continent) + National + Local	3	1	No documented connections	0	0	2.17
OpenStreet Map	PPS	International + National + Local	3	1	No documented connections	0	0	1.66
P2P Foundation	PPS	International + National + Local (Ghent)	3	1	Shared space: open knowledge commons cluster	1	0.33	2.41

Sensorica	PPS	National (Canada) + Local (Montreal)	2	0.67	Shared space: open knowledge commons cluster	1	0.33	1.92
Stop LAPD Spying	PPS	National + Local (Los Angeles)	2	0.67	Shared space: algorithmic accountability cluster	1	0.33	1.66
TIC A.C.	PPS	National (Mexico) + Local + Some international (Berlin)	2	0.67	Shared space: community network builders cluster	1	0.33	1.83
WAO	PPS	National (US) + Local	2	0.67	Shared space: algorithmic accountability cluster	1	0.33	1.83
WikiHouse	PPS	National (UK) + Some international (distributed fabricators)	2	0.67	Shared space: open knowledge commons cluster	1	0.33	1.66
Zenzeleni	PPS	International + National (SA) + Local (rural)	3	1	Shared space: community network builders cluster	1	0.33	2.58